



ELSEVIER

Journal of Non-Crystalline Solids 290 (2001) 25–31

JOURNAL OF
NON-CRYSTALLINE SOLIDS

www.elsevier.com/locate/jnoncrysol

Nucleation and crystallisation of transparent, erbium III-doped, oxyfluoride glass-ceramics

L.L. Kukkonen^a, I.M. Reaney^a, D. Furniss^a, M.G. Pellatt^b, A.B. Seddon^{a,*}^a Department of Engineering Materials, Centre for Glass Research, University of Sheffield, Sheffield S1 3JD, UK^b Merck Research and Development UK, Floor 4, Building 35, University of Southampton, Southampton S17 1BJ, UK

Received 16 April 2001

Abstract

A new Er^{3+} -doped oxyfluoride glass-ceramic composition is reported, based on published compositions but with improved glass stability on reheating. A series of heat treatments up to 168 h at 394°C, close to T_g (glass transformation temperature), has been carried out and yielded visually transparent materials. X-ray diffractometry (XRD) has shown that a fluorite-structured phase with lattice parameter 0.574 nm has crystallised after the 394°C/168 h heat treatment. High-resolution, transmission electron microscope (HRTEM) images of the as-annealed glass prior to heat treatment revealed that the glass appeared not to have phase separated on melt-quenching. However, after heat treatment for 6 h at 394°C, ordered regions (0.5 nm) which had a dark contrast with respect to the matrix were imaged. These regions are believed to be associated with onset of nucleation of the fluorite phase. The regions grew with heat treatment time at 394°C initially as approximately spherical crystals and at longer times becoming dendritic with rounded arms protruding from a central nucleus. After heat treatment of 0.25 h at 600°C, glasses appeared visually pink and opaque, being heavily crystallised. XRD showed that again the fluorite-structured phase had grown and HRTEM imaging indicated that the fluorite-type crystals had coalesced into comparatively large spheres of 203 ± 34 nm diameter in a glassy matrix. Energy dispersive spectroscopy (EDS) proved Er^{3+} had preferentially separated into the fluorite-structured phase, and that the latter phase is PbF_2 rather than a $\text{Pb}_x\text{Cd}_{[1-x]}\text{F}_2$ solid solution as previously supposed. © 2001 Elsevier Science B.V. All rights reserved.

1. Introduction

Rare-earth-doped, oxyfluoride glass-ceramics were reported in the 1970s [1]; they have recently become topical because of the current drive for novel glass waveguides for active signal processing

devices, including amplifiers, sensors and sources. Oxyfluoride glass-ceramics are accessed via melting of oxyfluoride glasses followed by heat treatment at, or above, the glass transformation region (T_g). Such heat treatment gives a secondary phase consisting of fluorite-structured crystallites in which, it has been suggested that, doped rare earth ions selectively partition [1–4]. The size of the crystalline domains can be sufficiently small not to cause excess light scattering and the glass-ceramics appear transparent to the naked eye. The great advantages of this approach are that: (i) doped rare earth ions reside in a low phonon energy crystalline environment and hence excited state lifetimes and cross-sections are

* Corresponding author. Present address: Centre for Advanced Materials, School of Mechanical, Materials, Manufacturing Engineering and Management, University of Nottingham, University Park, Nottingham NG7 2RD, UK. Tel.: +115-951 3740; fax: +44-9 115 951 3741.

E-mail address: angela.seddon@nottingham.ac.uk (A.B. Seddon).

potentially larger than in a glassy environment, yet (ii) the glass host matrix is silicate based with suitably good mechanical and chemical properties. Fabrication of waveguides is a matter of avoiding premature, uncontrolled phase separation which could occur during processing. Hence the challenge of making transparent glass-ceramic waveguides is to achieve controlled structuring of the glass at both the nanometre- and micrometre-scales.

Our overall aim is to pursue a deeper understanding of the mechanisms of phase separation in rare earth doped oxyfluoride glass-ceramics to help design methods for waveguide making. A new rare earth doped, glass-ceramic composition is reported here, based on previously published compositions [3,5]. Using the differential thermal analysis (DTA) data of the parent oxyfluoride glass as a guide to choice of temperature, a series of heat treatments has been carried out on the oxyfluoride glass-forming liquid. Following heat treatment, the quenched products have been investigated by means of X-ray diffractometry (XRD) and analytical, high-resolution transmission electron microscopy (HRTEM) with energy dispersive X-ray analysis (EDS). For the first time, ordered structures of about 0.5 nm have been imaged which are believed to be associated with nucleation of the fluorite-structured nano-crystalline phase. Direct analytical evidence is presented of the selective partitioning of Er^{3+} into the fluoride phase and that this latter phase is PbF_2 rather than a solid solution of $\text{Pb}_x\text{Cd}_{[1-x]}\text{F}_2$ as previously supposed [2,3].

2. Experimental

2.1. Glass preparation

Table 1 shows the batch compositions of the two types of oxyfluoride glasses prepared during this work. Glasses 1 and 2 were batched with 1 and

4 mol% ErF_3 , respectively; the former composition is due to Tick et al. [3]. The supplier, and purity, of each batch precursor was as follows: SiO_2 (Tilcon Industrial Minerals, Type L30A Loch Aline, 99.8%, Fe_2O_3 0.009%); Al_2O_3 (Alfa, 'Puratronic' 99.997%, 110 μm particle size); CdF_2 (Alfa, 99.99%); ZnF_2 (Merck, 'Fluorotran' 99.995%); PbF_2 (Merck, 'Fluorotran' 99.997%); YF_3 (Merck, 'Fluorotran' 99.99%) and ErF_3 (Merck, 'Fluorotran' 99.99%).

15 g batches were melted in a Pt/2 wt% Rh, 15 ml crucible, with a Pt lid, placed inside a covered silica liner (Heraeus) in a resistance furnace. Dried air (BOC) was flowed through the silica liner and the batch was introduced into the furnace at 1080°C and melted for 50 min. The transparent glass liquid was then cast in the ambient atmosphere into an aluminium mould. The thin plates of transparent glass so-produced were annealed at 380°C for 1 h, cooled in the furnace, then ground and polished to a 1 μm finish, and subsequently examined for crystallinity under an optical microscope (Polyvar Met).

2.2. DTA

Using DTA equipment (Perkin Elmer DTA7), samples were held in a Pt crucible and analysed against a calcined Al_2O_3 reference at a heating rate of 15°C min^{-1} , under flowing dried Ar (BOC). Characteristic temperatures taken from the DTA curves were: the extrapolated onset of T_g , and T_{conset} which was taken as the temperature at which the DTA curve first deviated from the baseline prior to the lowest temperature crystallisation peak above T_g .

2.3. Heat treatment

The annealed, polished glass plate samples were cut into smaller plates and a fresh sample was ta-

Table 1
Mol% batch compositions of glasses prepared

Glass	SiO_2	$\text{AlO}_{1.5}$	CdF_2	PbF_2	ZnF_2	YF_3	ErF_3
1 ^a	30	15	29	17	5	3	1
2	30	15	29	17	5	0	4

^a Composition due to Tick et al. [3].

ken for each heat-treatment run (see Table 3). The heat-treatment furnace (Johnson Matthey, serial number 4257) was calibrated and held at $394 \pm 1^\circ\text{C}$ under flowing, dried Ar (BOC). The glass sample was placed in a hand-made wire basket holder (80 Nikrothal, wire diameter 0.545 mm) attached to the end of the measuring thermocouple which was inserted just inside the furnace end-cap and left to equilibrate at 30°C under the Ar flow for 15 min before being introduced into the furnace hot zone. After heat treatment, the sample was withdrawn again to just inside the furnace end-cap to quench to 30°C . Samples were stored in a desiccator charged with silica-gel.

2.4. XRD

The heat treated, or as-annealed, glass plate samples were mounted directly on aluminium holders using adhesive (UHU© glue, mixed with acetone) and analysed by means of an X-ray diffractometer (Philips 1700) with $\text{CuK}\alpha$ radiation at 50 kV and 40 mA, between 10° and $70^\circ 2\theta$ with a step size of $0.02^\circ 2\theta$ and scan speed of $0.1^\circ 2\theta \text{ min}^{-1}$. Patterns were indexed using software (SIETRONICS: PW 1050 Diffractometer Automation Sie 122D).

2.5. TEM

Glass samples were ground on a glass slide to a thickness $<100 \mu\text{m}$ using SiC paper ending with 1200 grit. An epoxy adhesive was used to mount a copper support ring of diameter 3.05 mm and a 1 mm hole on the thinned sample. Thinning to perforation was achieved using ion beam milling equipment (Gatan Dual Ion Beam Miller) operated at an incident angle of 15° , an accelerating voltage of 6 kV and a combined gun current of 6

mA. Samples were carbon coated and analysed by means of TEM (JEOL JEM 3010, No. EMI 130005-8 at 300 kV with Analytical Link exL, Pentalet) using an ultra-thin window.

3. Results

3.1. Glass melting

The products of glass melting were crystal-free according to optical microscopy and were visually transparent appearing pink due to Er^{3+} . Characteristic temperatures from the DTA curves (Fig. 1) are given in Table 2. For Glass 2 (4 mol% ErF_3), the mean characteristic temperatures of DTA runs on four glasses of nominally the same batch composition, melted from different batches, are given: T_g was $394 \pm 6^\circ\text{C}$ and there were two crystallisation peaks at T_{conset} $475 \pm 7^\circ\text{C}$ and at above 600°C , respectively, where the higher temperature crystallisation peak may have comprised coalesced

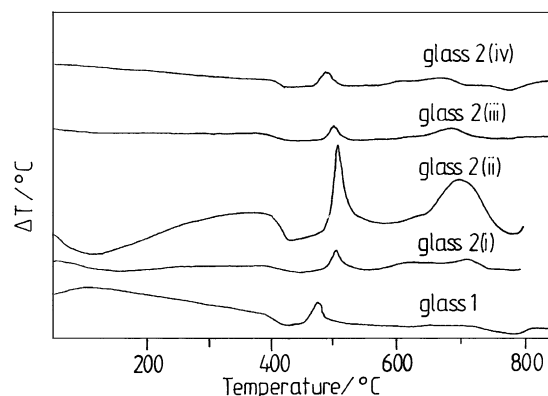


Fig. 1. Typical DTA curves of prepared oxyfluoride glasses. (See Table 1 for batch compositions of glasses and Table 2 for characteristic temperatures taken from the DTA curves.)

Table 2
Characteristic temperatures taken from DTA curves (Fig. 2)

Glass	Mol% ErF_3	T_g ($^\circ\text{C}$)	T_{conset} ($^\circ\text{C}$)	$\{T_{\text{conset}} - T_g\}$ ($^\circ\text{C}$)	T_{liq} ($^\circ\text{C}$)
1	1	396	434	38	
2	4	$^a 394 \pm 6$	$^a 475 \pm 7$	$^a 81 \pm 11$	$805^b \pm 6$

^a Average of DTA runs on four different glasses of the same nominal batch composition (Glasses 2(i)–(iv)).

^b Two runs only. See Table 1 for batch compositions of glasses.

peaks. These characteristic temperatures were taken as a guide to design suitable time/temperature schedules for the subsequent heat treatments. To capture the very earliest stages of crystal growth, a heat treatment temperature close to T_g of 394°C was selected for times from 0.5 to 168 h (Table 3). A fresh glass sample was taken for each heat-treatment run and samples were from the same or different batched melts of nominally the same composition.

3.2. Results of heat treatment at 394°C of Glass 2, batch composition (mol%): 30SiO₂–15AlO_{1.5}–29CdF₂–17PbF₂–5ZnF₂–4ErF₃

3.2.1. Visual appearance

All of the heat treatments at 394°C from 0.5 to 168 h of Glass 2 samples, (batch mol%: 30SiO₂–15AlO_{1.5}–29CdF₂–17PbF₂–5ZnF₂–4ErF₃) yielded materials which were crystal-free according to optical microscopy and were visually transparent, appearing pink due to Er³⁺ (Table 3).

3.2.2. XRD

Fig. 2 shows the XRD patterns for samples of Glass 2 both as-annealed and after heating at 394°C for times up to 168 h. After 0.5–16 h heat treatment at 394°C, the XRD patterns showed two amorphous halos which tended to become decreased in width and increased in counts with time.

Table 3

Heat treatment schedules of Glass 2 samples (batch mol%: 30SiO₂–15AlO_{1.5}–29CdF₂–17PbF₂–5ZnF₂–4ErF₃; glasses 2(i)–(iii) were melted separately) together with visual appearance of samples following the heat treatment

Sample	Temperature (°C)	Time (h)	Visual appearance
2(i)	394 ± 1.8	0.5	Transparent pink
2(i)	394 ± 0.3	1.0	Transparent pink
2(i)	394 ± 0.5	3.0	Transparent pink
2(i)	394 ± 2.4	6.0	Transparent pink
2(ii)	394 ± 0.9	6.0	Transparent pink
2(ii)	394 ± 0.3	16.0	Transparent pink
2(ii)	394 ± 1.0	48.0	Transparent pink
2(iii)	394 ± 2.5	72.0	Transparent pink
2(iii)	394 ± 3.5	96.0	Transparent pink
2(iii)	394 ± 3.0	120	Transparent pink
2(iii)	394 ± 3.0	144	Transparent pink
2(iii)	394 ± 3.5	168	Transparent pink
2(iii)	600 ± 1.4	0.25	Opaque pink

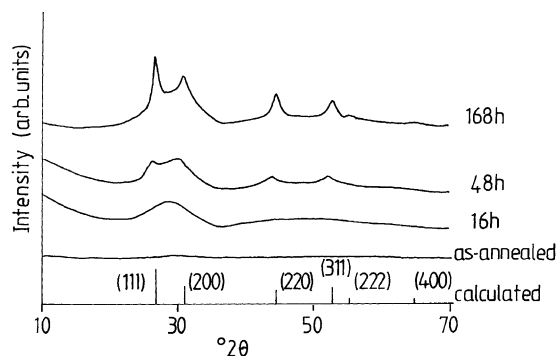


Fig. 2. X-ray diffraction patterns of Glass 2 (batch mol%: 30SiO₂–15AlO_{1.5}–29CdF₂–17PbF₂–5ZnF₂–4ErF₃) of as-annealed glass and samples heated at 394°C for 16 h (Glass 2(ii)), 48 h (Glass 2(ii)) and 168 h (Glass 2(iii)) and a calculated pattern assuming that the developing crystals have a face-centred cubic structure and lattice parameter of 0.574 nm.

After 48 h heat treatment at 394°C, several small diffraction peaks had developed whose widths decreased and counts increased at longer times at 394°C.

3.2.3. HRTEM and EDS

High-resolution transmission electron micrographs of the as-annealed glass showed a uniform field with no visible structuring (Fig. 3(a)). EDS of the as-annealed glass (Fig. 3(b)) had peaks due to Si, Al, Zn, Cd, Pb and Er as expected.

However, after heat treatment for 6 h at 394°C HRTEM showed ordered regions of dark contrast, of 0.5 nm diameter and embedded in a glass matrix. Based on their dark contrast with respect to the matrix and evidence of ordering it is suggested that they are associated with nucleation of the fluorite-structured crystalline phase (Fig. 4). The regions grew with heat treatment time at 394°C initially as approximately spherical crystals. After 168 h heat treatment at 394°C, the crystals exhibited the onset of becoming dendritic with rounded arms protruding from a central nucleus of overall diameter 17.9 ± 2.7 nm. EDS results were ambiguous and tended to reflect the composition of the glass matrix.

After heat treatment at 600°C/0.25 h, glasses appeared visually pink and opaque, being heavily crystallised. XRD showed that again the fluorite phase had grown (lattice parameter 0.575) and

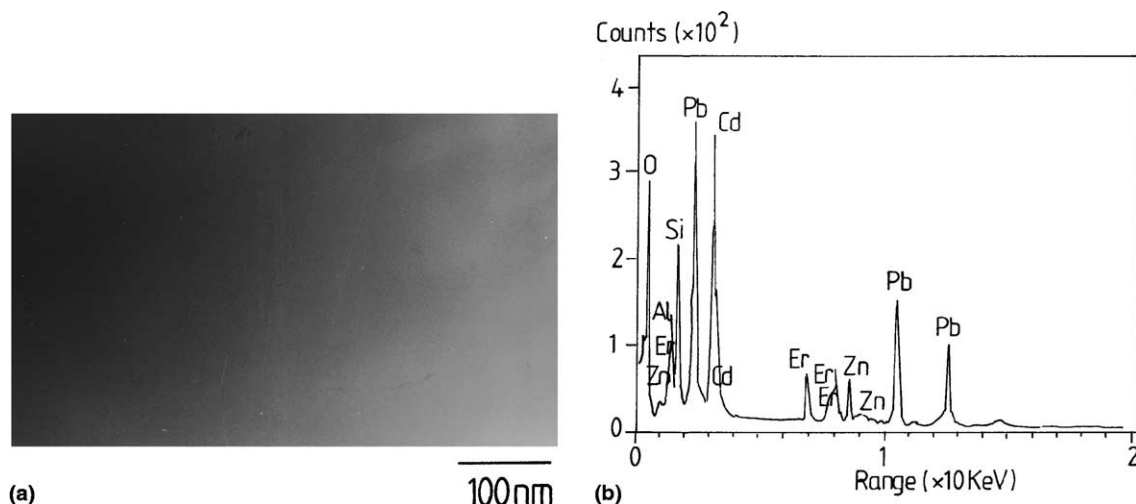


Fig. 3. As-annealed Glass 2 (iii) (batch mol%: 30SiO₂–15AlO_{1.5}–29CdF₂–17PbF₂–5ZnF₂–4ErF₃): (a) high-resolution transmission electron micrograph, (b) energy dispersive spectrum (of image in (a)).

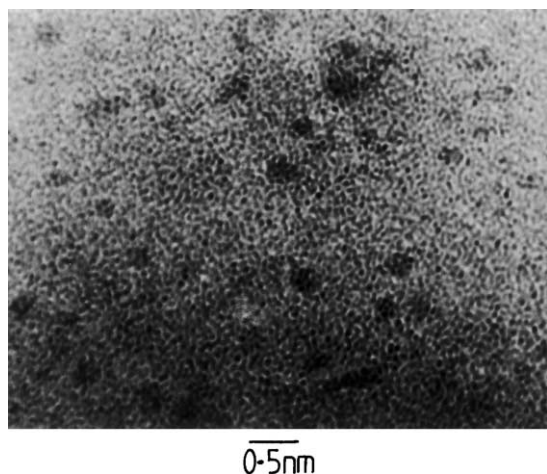


Fig. 4. High-resolution transmission electron micrograph of Glass 2 (i) (batch mol%: 30SiO₂–15AlO_{1.5}–29CdF₂–17PbF₂–5ZnF₂–4ErF₃) heat treated at 394°C/6 h.

HRTEM imaging indicated that the fluorite crystals had coalesced into comparatively large spheres of diameter 203 ± 34 nm diameter in a glassy matrix (Fig. 5(a)). EDS spectra of the crystals and glassy matrix are shown in Figs. 5(b) and (c). The EDS spectrum of the crystalline phase is Cd deficient but rich in Er and Pb. No peaks of zinc or aluminium can be observed in the spectrum. The EDS spectrum of the glass matrix shows low peak

intensity of Er, Zn and Al, and the Pb and Cd peaks are relatively strong. The Si peak is intense in each spectrum. However it is believed that the relative intensity of this peak is unreliable since it is possibly due to an artefact associated with contamination from the glue used while preparing the TEM samples.

In summary, from all of the evidence, the fluorite-structured nano-crystals are Pb and Er rich and the glass matrix deficient in Pb and Er.

4. Discussion

From Fig. 1 and Table 2, the Glass 2 composition, which was newly developed here, which contained 4 mol% ErF₃, as-batched, can have a ($T_{\text{conset}} - T_g$) gap of $81 \pm 11^\circ\text{C}$ compared to a gap of 38°C for Glass 1, batched with 1 mol% ErF₃. The batch-to-batch variation for nominally the same glass composition lies partly in the difficulty of assessing T_{conset} but may be due to variation in amount of volatilisation during the glass melting [6].

The ($T_{\text{conset}} - T_g$) temperature gap is often taken as a simple measure of glass stability on reheating. Glass shaping usually must either take place in this gap or during quenching from the liquidus. A larger gap can imply less tendency for

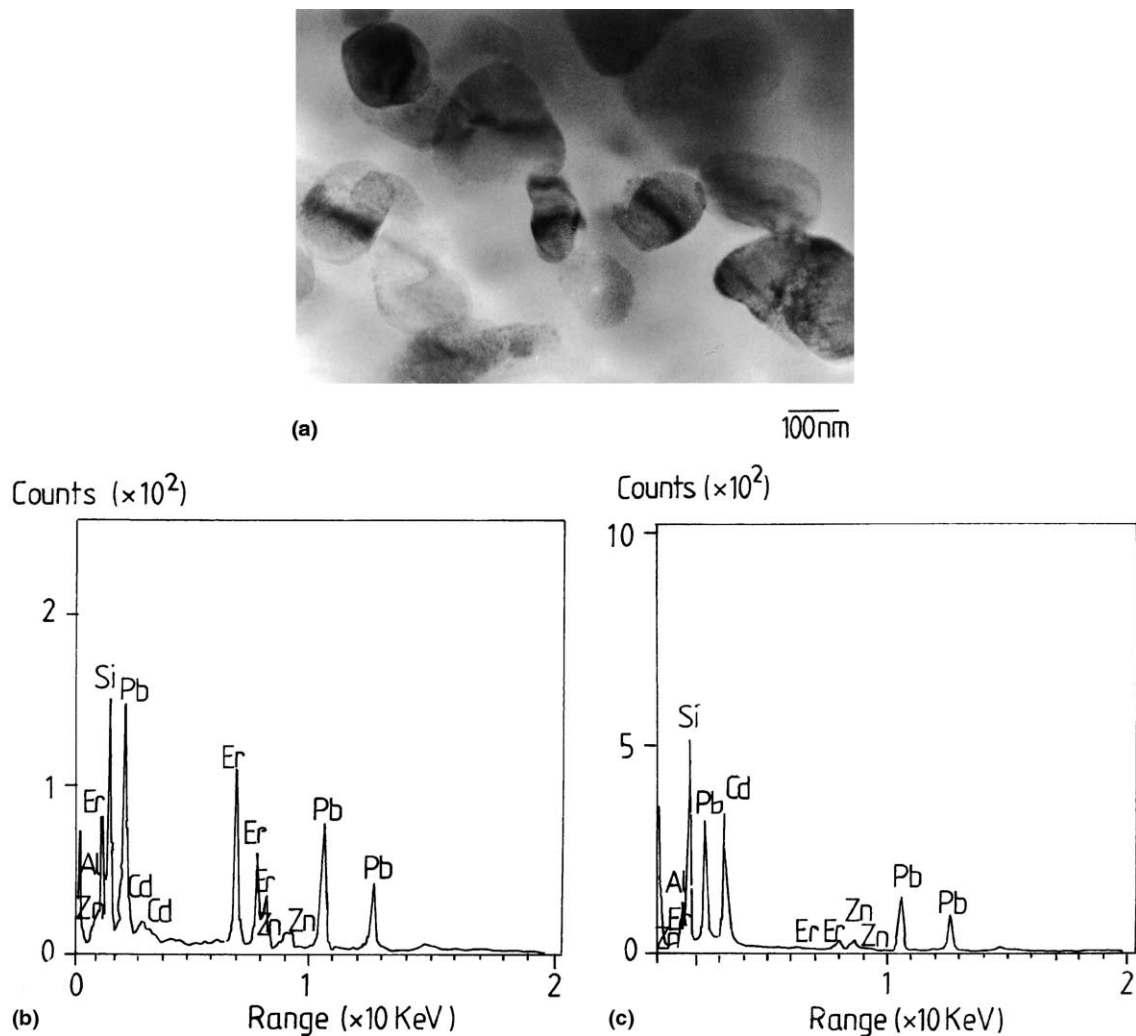


Fig. 5. Glass 2 (ii) (batch mol%: 30SiO₂–15AlO_{1.5}–29CdF₂–17PbF₂–5ZnF₂–4ErF₃) heat treated at 600°C/0.25 h: (a) high-resolution transmission electron micrograph, (b) energy dispersive spectrum of spherical crystals (imaged in (a)), (c) energy dispersive spectrum of residual glass matrix (imaged in (a)).

devitrification during shaping on reheating but suitable viscosities, appropriate for the shaping process, must be accessible in the gap.

Fig. 3(a) is a typical high-resolution micrograph of the as-annealed Glass 2 (batched with 4 mol% ErF₃). The lack of crystalline features and the homogeneity of the image typically observed is *prima facie* evidence for this glass not having undergone phase separation during quenching of the liquid nor during annealing. The annealing schedule involved 1 h at 380°C, then cooling in the furnace by

switching off the furnace. From HRTEM imaging there was in fact no phase separation until after 6 h heat treatment at 394°C, after which 0.5 nm ordered structures were evident (Fig. 4). We propose that these are early stages in the nucleation of the fluorite-structured crystalline phase.

The XRD patterns of Glass 2 samples (batched with 4 mol% ErF₃) heat treated from 0.5 to 168 h at 394°C (Fig. 2) showed similar peak development to that observed by Wang and Ohwaki [2] and Tick et al. [3] for similar, but not identical, glasses

which they claimed was due to growth of $\text{Pb}_x\text{Cd}_{1-x}\text{F}_2$ in which Er^{3+} had partitioned. The calculated pattern in Fig. 2 assumes that the developing crystals have a face-centred cubic structure with a lattice parameter of 0.574 nm. The XRD pattern of a Glass 2 sample heat treated at 600°C for 0.25 h showed the same peak development and hence growth of the fluorite-structured phase; the calculated lattice parameter was 0.575 nm. Small, but non-systematic, differences in the lattice parameter of the fluorite-structured phase were observed after different heat treatments which is likely to have been due to slight daily variation in the operating conditions of the diffractometer which was not calibrated.

Irrefutable evidence that erbium III ions have indeed preferentially partitioned to the separating fluoride nano-crystalline phase may be had by comparing the increased intensity of the EDS peaks due to Er^{3+} observed for the approximately spherical crystals imaged in the sample compared to the residual glass matrix heat treated at 600°C/0.25 h (Fig. 5). Moreover, the ratio of EDS peak intensities due to Er^{3+} ions relative to the other species in the fluorite-structured crystalline phase is greater than that in the original as-annealed glass (compare Fig. 5(b) with Fig. 3(b)). Finally, we have proved for the first time that this type of oxyfluoride glass phase separates to a mainly PbF_2 crystalline phase hosting the Er^{3+} dopant (Fig. 5(b)), and not a more complex solid solution of CdF_2 with PbF_2 , i.e., $\text{Pb}_x\text{Cd}_{1-x}\text{F}_2$, as previously supposed [2,3].

5. Conclusions

1. A new erbium III-doped oxyfluoride glass-ceramic has been developed with improved ($T_{\text{onset}}-T_g$) gap compared to related compositions [2,3].
2. XRD confirmed that a similar fluorite-structured secondary phase to those reported by oth-

ers [2,3] grows on heat treatment at times of up to 168 h at 394°C (close to T_g) and after heat treatment of 0.25 h at 600°C.

3. HRTEM has indicated the absence of crystallisation in the as-annealed glass.
4. HRTEM has revealed that after 6 h at 394°C, ordered structures of about 0.5 nm diameter have grown and it is proposed that these are the first step in the development of nucleation of the fluorite-structured crystalline phase.
5. HRTEM/EDS proves that Er^{3+} preferentially partitions to the fluorite-structured crystalline phase and that this phase is PbF_2 and not a complex $\text{Pb}_x\text{Cd}_{1-x}\text{F}_2$ solid solution as previously supposed.

Acknowledgements

The authors are grateful to Merck Ltd. and the Engineering and Physical Science Research Council for support for L.L.K. in the form of an EPSRC CASE PhD studentship.

References

- [1] F. Auzel, D. Pecile, D. Morin, J. Electrochem. Soc. 122 (1975) 101.
- [2] Y. Wang, J. Ohwaki, Appl. Phys. Lett. 63 (24) (1993) 3268.
- [3] P.A. Tick, N.F. Borelli, L.K. Cornelius, M.A. Newhouse, J. Appl. Phys. 78 (11) (1995) 6367.
- [4] P.A. Tick, Ultra-transparent glass-ceramics for photonic applications, invited paper, in: Proceedings of 11th International Symposium Non-Oxide and New Optical Glasses, Sheffield University, 1998, Chair: A.B. Seddon, published by the Society of Glass Technology, Sheffield, UK.
- [5] L.L. Kukkonen, D. Furniss, I.M. Reaney, A.B. Seddon, Transparent, rare earth doped, oxyfluoride glass-ceramics for photonics, in: Proceedings of Photons, Glasses and Coatings, Society of Glass Technology, April 1999, Pilkington Technology Centre, Lathom, UK.
- [6] M. Braglia, C. Bruschi, G. Dai, J. Kraus, S. Mosso, M. Baricco, L. Battezzati, F. Rossi, J. Non-Cryst. Solids 256&257 (1999) 170.